

MILESTONE CONTRACT

ARCH-001 V1.0 (BLUEPRINT)

3-TIER HIGH-AVAILABILITY



Module 8: Hardware Integration, QR Handheld Devices, ESC/POS Printers & 3-Tier Server Setup

Comprehensive 3-week milestone agreement, dedicated infrastructure engineering team breakdown, 3-tier server cluster deployment (Dev / Staging / Prod), industrial Android PDA barcode scanner listeners, ESC/POS thermal printing daemons, and biometric clock integrations based on verified Technical Architecture Blueprint ARCH-001 v1.0 for Popular Auto Spare & A/C Parts Co. W.L.L.

TOTAL MODULE BUDGET

BD

1,142.727

PROJECT TIMELINE

3 Wks (0.75 Mo)

WEEKLY RESOURCE RATE

BD 380.909

PAYMENT TRANCHES

3 Gates

(33.3% Ea)

Section Module 8: Functional Scope & Architecture

2.0 Breakdown (BA Ref: ARCH-001)

FULL SPECIFICATION MAPPING

Derived directly from **ARCH-001: Technical Architecture & Hardware Infrastructure Blueprint**, the implementation scope is structured across 5 core technical pillars:

2.1 3-Tier Enterprise Server Cluster & Virtualization

- **Environment Isolation:** Provisioning of isolated Production, Staging, and Development server instances with Linux OS kernel hardening.
- **Database Clustering:** High-performance MariaDB Enterprise database engine tuning (InnoDB buffer pool, query cache, SSD transaction log optimization).
- **Web & App Stack:** Nginx reverse proxy, PHP 8.x-FPM thread pools, SSL/TLS Let's Encrypt automated certificate renewals, and HTTP/2 acceleration.
- **High Availability & Snapshots:** Automated nightly snapshot replication with incremental data sync to off-site SAN backup repositories.

2.2 Industrial Handheld PDA & Barcode Scanner Daemon

- **Android PDA Scanner Engine:** Dedicated scanning service listener integrated directly into Zebra / Chainway / Urovo industrial Android handhelds.
- **Real-Time PCode Lookup:** Instant WebSocket and REST API inventory lookups upon 1D/2D QR code scans with sub-100ms response time.
- **Warehouse Picking Verification:** Visual and acoustic feedback on PDA screens for bin location validation, batch selection, and stock discrepancy alerts.
- **Offline Buffer Mode:** Local SQLite buffer on PDA devices to capture stock verification counts during temporary branch Wi-Fi disconnections.

2.3 ESC/POS Thermal Receipt & Industrial Zebra Label Daemon

- **Direct Network ESC/POS Driver:** Driverless raw TCP socket printing over port 9100 for high-speed POS thermal receipt printers across all branch counters.

- **Automatic Cash Drawer Kick:** Standard ESC/POS pulse codes triggered upon cash transaction completion and payment confirmation.
- **ZPL Industrial Label Printing:** Zebra Programming Language (ZPL) label generator for warehouse rack bin tags, PCode QR stickers, and bulk carton labels.
- **Multi-Branch Spooler:** Centralized printing queue management with automated retry on paper out, cover open, or network disconnection.

2.4 Biometric Clock Network Fleet & Push Sync Listener

- **Multi-Branch Clock Network:** TCP/IP network listener service communicating with ZKTeco biometric attendance terminals across all corporate offices and branches.
- **Real-Time Event Stream:** Automated instant push of employee punch logs (IN/OUT, Break, Overtime) into Module 7 HRMS attendance staging tables.
- **Heartbeat Health Monitoring:** Automated ping monitoring to detect offline terminals and send alert notifications to IT administrators.
- **Biometric User Enrollment Sync:** Centralized employee ID and biometric template synchronization from HRMS directly to remote clock devices.

2.5 Infrastructure Security, VLAN Segmentation & SAN Backup Automation

- **Network Security:** Dedicated VLAN segmentation isolating POS terminal subnets, PDA scanner Wi-Fi subnets, and database backend infrastructure.
- **Firewall Hardening:** Strict UFW / iptables rules restricting database and SSH access to whitelisted static corporate branch IP addresses.
- **Automated SAN Backup:** Automated cron pipeline taking daily encrypted MySQL dumps and rsync file store snapshots to a secondary off-site backup vault.
- **Failover & Disaster Recovery:** Documented Standard Operating Procedures (SOP) with Recovery Point Objective (RPO < 4 hours) and Recovery Time Objective (RTO < 2 hours).

Section Project Team Structure & Dedicated Resource

3.0 Pricing (3 Weeks / 0.75 Months)

COST & ALLOCATION BREAKDOWN

To deploy the enterprise 3–tier server infrastructure and integrate all physical hardware peripherals seamlessly, SaNDS Lab allocates a specialized 3–member engineering team:

ROLE & EXPERTISE	FTE ALLOCATION	WEEKLY RATE	3-WEEK TOTAL (BD)	PRIMARY DELIVERABLES
Principal Cloud & Infrastructure Architect	0.50 FTE	BD 152.364	BD 457.092	3–tier server cluster deployment, Linux OS hardening, MariaDB database clustering & tuning
Embedded Hardware & Peripherals Engineer	1.00 FTE	BD 152.364	BD 457.092	Android PDA scanner daemon, ESC/POS raw socket printing driver, ZPL label generator
Network Security & DevOps Specialist	0.50 FTE	BD 76.181	BD 228.543	Biometric clock push listener, VLAN segmentation, firewall rules, automated SAN backup scripts

ROLE & EXPERTISE	FTE ALLOCATION	WEEKLY RATE	3-WEEK TOTAL (BD)	PRIMARY DELIVERABLES
Total Dedicated Team Allocation	2.00 FTE	BD 380.909 / Wk	BD 1,142.727	3 Working Weeks (15 Days)

Section 4.0

Detailed Milestone Breakdown & Deliverables (3-Week Roadmap)

SPRINT EXECUTION PLAN

The implementation is organized into 3 sequential weekly milestones (33.33% each), guaranteeing rapid deployment, verified hardware testing, and flawless go-live integration:

- 1

Milestone 8.1: 3-Tier Server Virtualization, Database Clustering & Security Hardening

WEEK 1 (1 WK)

Setup of isolated Production, Staging, and Development server environments, MariaDB high-performance configuration, Nginx reverse proxy with SSL/TLS, and Linux OS firewall security hardening.

✓ 3-Tier Production, Staging & Dev Instances Active

✓ MariaDB Enterprise Clustering & Buffer Tuning

✓ Nginx SSL/TLS Reverse Proxy & HTTP/2 Configured

✓ Firewall Hardening & IP Whitelist Gateways
- 2

Milestone 8.2: Industrial PDA Scanner Listener & ESC/POS Thermal / Zebra Label Daemon

WEEK 2 (1 WK)

Integration of Android handheld PDA 2D barcode scanner engine with ERP stock verification APIs, driverless TCP 9100 ESC/POS thermal receipt printing, cash drawer kick, and ZPL barcode label printing.

- ✓ Android PDA 2D Barcode Scanning Receiver
- ✓ Sub-100ms PCode Real-Time Stock Lookup API
- ✓ ESC/POS Network Receipt Printing & Cash Drawer Trigger
- ✓ Zebra ZPL Barcode Rack Label Generator Active

3

Milestone 8.3: Biometric Clocks Multi-Branch Sync, SAN Backup Automation & Go-Live Acceptance

WEEK 3 (1 WK)

Multi-branch ZKTeco biometric clock TCP sync daemon deployment, automated off-site encrypted SAN backups, disaster recovery failover drill, and executive infrastructure sign-off.

- ✓ ZKTeco Biometric Clock Multi-Branch Network Sync
- ✓ Daily Encrypted SAN Backup & Snapshot Pipeline
- ✓ Disaster Recovery Drill (RTO < 2h, RPO < 4h)
- ✓ Final Production Go-Live Hardware Acceptance

Section 5.0 Milestone Payment Structure & Invoicing Schedule (3-Week Model)

VERIFIABLE MILESTONE BILLING

Invoicing is strictly tied to verifiable technical deliverables and stakeholder sign-off across 3 equal weekly milestone gates:

GATE	MILESTONE DELIVERABLE FOCUS	TIMELINE	% FEE	INVOICE AMOUNT (BD)	VERIFICATION CRITERIA
Gate 8.1	3-Tier Server Virtualization & Database Clustering	End of Wk 1	33.33%	BD 380.909	Dev/Staging/Prod instances online & SSL verified
Gate 8.2	PDA Scanner Daemon & ESC/POS / Zebra Label Printing	End of Wk 2	33.33%	BD 380.909	PDA PCode barcode scan & receipt print verified
Gate 8.3	Biometric Clock Sync, SAN Backup & Final Go- Live	End of Wk 3	33.34%	BD 380.909	Biometric punch push verified & backup drill approved
Total Module 8 Professional Investment			100.00%	BD 1,142.727	3 Equal Milestone Tranches

Section 6.0 Working Calendar, SLA, 15-Day Grace Period & Legal Terms

GOVERNANCE FRAMEWORK

Bahrain Working Calendar & Schedule

Standard working schedule is Sunday through Thursday (5 business days / 40 hours per week). Official Kingdom of Bahrain statutory public holidays are observed and accounted for in the 3-week timeline.

15-Day Quality Review & Grace Period

Following completion of each milestone gate, Popular Auto Parts is granted a 15–calendar-day review window to validate deliverables. If no written defects are reported within 15 days, milestone acceptance is confirmed.

Section 7.0 Stakeholder Authorization & Digital Sign-Off Console

EXECUTIVE ACCEPTANCE

By electronically signing below, the authorized executive stakeholders formally accept the scope, deliverables, pricing, and milestone schedule for **Module 8: Hardware Integration, QR Handheld Devices, ESC/POS Printers & Server Setup**.

Implementation Partner

 SIGNED

 PENDING SIGNATURE

SaNDS Lab / UniGlobal Consulting

 SaNDS Signature  Signature Pending

Signer:

Title:

Signed Date:

 Sign for SaNDS Lab

Client Organization

 SIGNED

 PENDING SIGNATURE

Popular Auto Spare & A/C Parts Co. W.L.L.



Client Signature



Signature Pending

Signer:

Title:

Signed Date:



Sign for Popular Auto Parts



Finalize & Lock Milestone Acceptance

✔ Document Accepted and Finalized into Master ERP Registry.